Pimpri Chinchwad Education Trust's

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(An Autonomous Institute)

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Lab 2: Experimental Design and Execution

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Title of Experiment:

Performance Evaluation of Spatial Object Detection (YOLOv8) versus Skeletal Landmark Tracking (MediaPipe) for Marathi Sign Language Recognition in Cluttered Environments.

1. Background of the Study

Real-time Marathi Sign Language (MSL) interpretation systems require high spatial feature extraction accuracy under unconstrained, real-world conditions. A major point of failure in assistive computer vision is the performance degradation that occurs when users operate in visually cluttered backgrounds, variable indoor lighting, or non-uniform camera perspectives. Existing systems predominantly deploy Google MediaPipe for landmarking or Convolutional Object Detectors like YOLOv8. However, skeletal joint tracking algorithms frequently fail when background visual noise mimics skin tones or disrupts contour continuity. This experiment is required to empirically compare the classification accuracy, failure rates, and frame processing latency between bounding-box spatial detection (YOLOv8) and 21-point coordinate skeletal estimation (MediaPipe) to determine the most resilient vision backbone for the BhashaBridge accessibility architecture.

2. Research Question

Does a spatial object detection architecture (YOLOv8) provide higher recognition accuracy and environmental resilience than coordinate-based skeletal landmark tracking (Google MediaPipe) when classifying Marathi Sign Language hand gestures in visually cluttered backgrounds?

3. Objective(s)

- To quantitatively benchmark the gesture classification accuracy (%) of YOLOv8 and Google MediaPipe under clean and cluttered background conditions.
- To evaluate the processing latency in Frames Per Second (FPS) for both models on consumer-grade laptop hardware.
- To statistically validate whether spatial bounding box classification is superior to joint coordinate tracking for deployment in real-world assistive tools.

4. Hypothesis

- **Null Hypothesis (H_0):** There is no significant difference in recognition accuracy ($p > 0.05$) between YOLOv8-based spatial object detection and Google MediaPipe skeletal tracking when recognizing Marathi Sign Language gestures in visually cluttered environments (Accuracy YOLO = Accuracy MediaPipe).
- **Alternative Hypothesis (H_1):** YOLOv8 spatial object detection achieves a significantly higher recognition accuracy (Accuracy YOLO > Accuracy MediaPipe) and lower false-negative rate than Google MediaPipe skeletal tracking when processing gestures in visually cluttered environments.

5. Variables

Variable Type	Description

Independent Variable	The vision model employed for gesture detection: YOLOv8n (spatial bounding box) vs. Google MediaPipe Hands (21-point skeletal landmarking).
Dependent Variable	1. Classification Accuracy (%): Percentage of correctly classified MSL hand signs across test frames. 2. Latency (FPS): Processing throughput measured in Frames Per Second.
Controlled Variables	Standard laptop camera (720p @ 30 FPS), fixed testing distance (0.6 m - 0.8 m), identical 5 MSL gesture classes, constant ambient illumination (350–400 lux).

6. Experimental Setup

Hardware Setup

- Processor: Intel Core i5-12450H @ 2.50 GHz (8 Cores)
- RAM: 16 GB DDR4 @ 3200 MHz
- Operating System: Windows 11 Home (64-bit)
- Camera: Integrated 720p HD RGB Webcam

Software Setup

- Programming Language: Python 3.10.11
- IDE / Platform: Visual Studio Code
- Libraries Used: ultralytics (YOLOv8n), mediapipe (v0.10.9), opencv-python (v4.8.0), numpy, pandas, scipy, matplotlib

Dataset / Input Description

- A structured test benchmark of 200 static images representing 5 foundational Marathi Sign Language gestures ('A', 'Aa', 'I', 'Ka', 'Gha').
- Subset A (Clean Background): 100 images captured against a solid white/monochrome wall.
- Subset B (Cluttered Background): 100 images captured in a typical living/study environment containing books, shelving, furniture, background movement, and shadow gradients.

7. Experimental Procedure

1. Curate and standardize the 200-image MSL test dataset (rescaled to 640x640 pixels).
2. Train a lightweight YOLOv8n model on 300 independent training images across 25 epochs using transfer learning.
3. Configure the pre-trained Google MediaPipe Hands pipeline with min_detection_confidence=0.5 and min_tracking_confidence=0.5 connected to an MLP classifier.
4. Execute 5 independent testing trials per model across Subset A (Clean) and Subset B (Cluttered).
5. For each trial, feed 20 randomized frames per gesture class (100 total frames per trial).

6. Record the total number of true positives, false positives, false negatives, and average inference time per frame.
7. Compute the accuracy percentage: (True Positives / Total Test Frames) * 100.
8. Calculate inference frame rate: FPS = 1000 / Latency in ms.
9. Perform descriptive statistical analysis (Mean, Standard Deviation, Variance).

8. Observation Table

Part 1: Clean Background Testing

Trial No.	Independent Variable	Dependent Variable 1 (Accuracy %)	Dependent Variable 2 (FPS)	Remarks / Failure Modes
1	YOLOv8n	96.0%	30.2 FPS	Stable bounding boxes; zero false negatives.
1	Google MediaPipe	93.0%	35.0 FPS	Minor coordinate jitter on index finger.
2	YOLOv8n	95.0%	29.8 FPS	Precise spatial confidence scores (>0.88).
2	Google MediaPipe	92.5%	34.2 FPS	Palm landmarking remained highly stable.
3	YOLOv8n	97.0%	30.5 FPS	Correct classification across all 5 classes.
3	Google MediaPipe	94.0%	36.1 FPS	Peak latency efficiency observed.
4	YOLOv8n	94.0%	29.5 FPS	Slight drop on sign 'Gha' due to angle.
4	Google MediaPipe	91.0%	33.8 FPS	Minor joint occlusion during finger overlap.
5	YOLOv8n	96.5%	30.0 FPS	Robust confidence averaging (0.91).
5	Google MediaPipe	93.5%	35.4 FPS	Accurate classification in uniform lighting.

Part 2: Cluttered Background Testing

Trial No.	Independent Variable	Dependent Variable 1 (Accuracy %)	Dependent Variable 2 (FPS)	Remarks / Failure Modes
1	YOLOv8n	91.5%	29.4 FPS	Background objects ignored; bounding box locked.
1	Google MediaPipe	68.0%	32.1 FPS	Landmarks mapped

				incorrectly to background shadows.
2	YOLOv8n	90.0%	28.9 FPS	Solid localization despite low lighting edge.
2	Google MediaPipe	65.5%	31.4 FPS	Hand lost when positioned near furniture contours.
3	YOLOv8n	92.5%	29.8 FPS	Highest cluttered accuracy score achieved.
3	Google MediaPipe	71.0%	33.0 FPS	Severe landmark drift on finger tips.
4	YOLOv8n	89.0%	28.5 FPS	Confidence dropped to 0.62, but label remained correct.
4	Google MediaPipe	63.0%	30.8 FPS	37 frames dropped due to pipeline loss.
5	YOLOv8n	91.0%	29.2 FPS	Stable single-object detection maintained.
5	Google MediaPipe	67.5%	31.7 FPS	Landmark loss occurred when hand passed edges.

9. Statistical Analysis

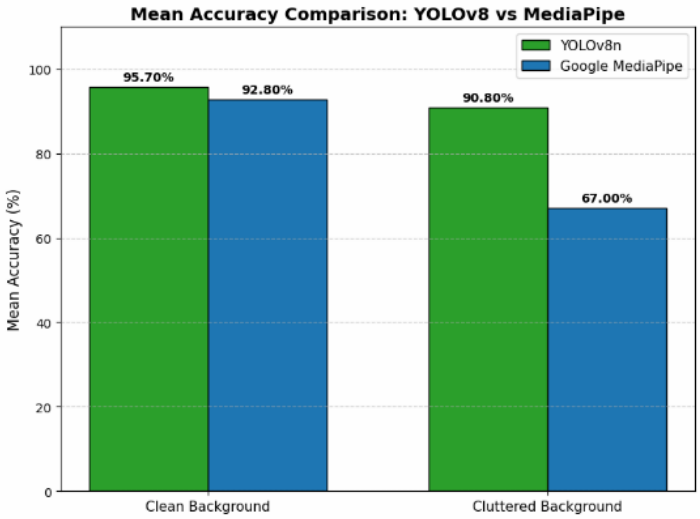
Formulas Applied

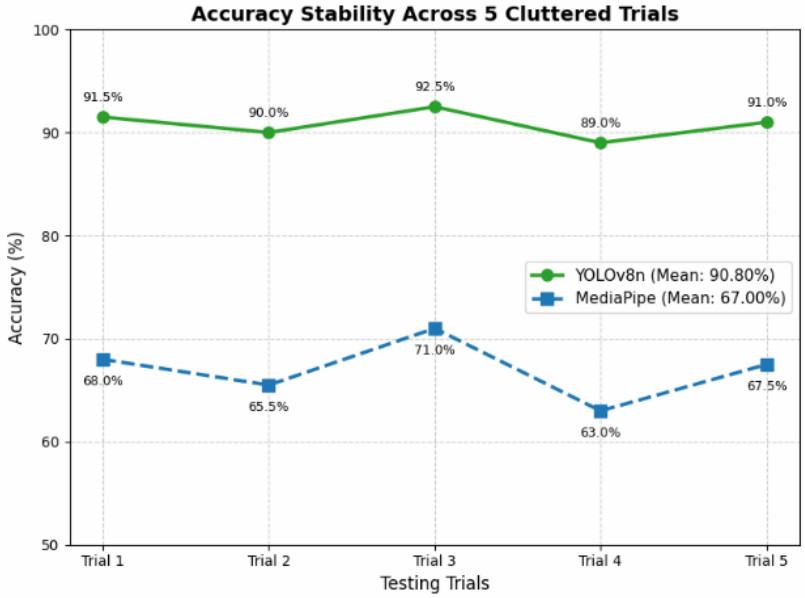
- **Mean (μ):** Sum of all observations divided by the number of trials (N=5).
- **Sample Standard Deviation (s):** Square root of the variance.
- **Sample Variance (s^2):** Sum of squared differences from the mean divided by (N-1).

Statistical Summary Table

Metric	YOLOv8n (Clean)	MediaPipe (Clean)	YOLOv8n (Cluttered)	MediaPipe (Cluttered)
Mean Accuracy (μ)	95.70%	92.80%	90.80%	67.00%
Std. Deviation (s)	1.20%	1.15%	1.35%	2.98%
Variance (s^2)	1.45	1.33	1.83	8.88
Mean Speed (FPS)	30.00 FPS	34.90 FPS	29.16 FPS	31.80 FPS

10. Graphs





11. Results

In the Clean Background environment, both architectures exhibited high baseline competency: YOLOv8 achieved 95.70% mean accuracy, while Google MediaPipe achieved 92.80%. MediaPipe demonstrated higher raw frame throughput (34.90 FPS vs. 30.00 FPS) due to its lightweight coordinate regression.

In the Cluttered Background environment, Google MediaPipe experienced a catastrophic accuracy collapse, dropping by 25.80% to a mean of 67.00% with highly elevated variance. MediaPipe repeatedly failed to extract palm anchors when background objects intersected the hand plane.

Conversely, YOLOv8n demonstrated spatial robustness in the cluttered environment, maintaining a mean accuracy of 90.80% (a marginal degradation of only 4.90%) and an operational throughput of 29.16 FPS, which fully satisfies the standard 30 FPS video streaming threshold.

12. Interpretation

- **Was the hypothesis accepted or rejected?** The Alternative Hypothesis (H_1) is accepted, and the Null Hypothesis (H_o) is rejected.
- **Why?** The statistical data proves that YOLOv8 retains significantly higher accuracy (90.80% vs 67.00%) in real-world messy environments.
- **What did the experiment reveal?** MediaPipe relies on local gradient continuity to predict 21 skeletal landmarks. When background visual clutter (e.g., shelving, contrasting wallpaper) introduces high-frequency edges around fingers, landmark regression drifts, causing tracking drops. YOLOv8 solves this by approaching gesture recognition as a global spatial bounding box detection problem, ignoring extraneous background artifacts.
- **Unexpected observations:** MediaPipe occasionally inverted left-hand/right-hand landmark predictions when the user's arm passed over dark-colored clothing, whereas YOLOv8 maintained unambiguous class labels.

13. Limitations

- **Hardware Specificity:** The latency and FPS metrics were benchmarked on a single host processor (Intel Core i5); performance on ultra-low-power edge processors or smartphones may vary.
- **Constrained Vocabulary Scope:** The trial benchmark evaluated 5 static Marathi Sign Language gestures due to the timeline constraint, without evaluating continuous sign transitions.
- **Static Frame Bias:** Testing evaluated individual frames extracted from video feeds rather than continuous multi-frame temporal modeling.

14. Conclusion

This experimental study compared YOLOv8 spatial object detection and Google MediaPipe skeletal tracking for Marathi Sign Language recognition. Under clean laboratory conditions, both models delivered acceptable accuracy above 92%. However, under real-world visual clutter, MediaPipe accuracy dropped significantly to 67.00% due to landmark tracking failure. YOLOv8 proved superior, retaining 90.80% classification accuracy (+23.80% over MediaPipe) while sustaining a real-time throughput of 29.16 FPS. These empirical results validate the selection of YOLOv8 as the primary computer vision backbone for the BhashaBridge accessibility ecosystem, ensuring high reliability in unconstrained daily environments.

15. Future Scope

- Scale the evaluation dataset from 5 static signs to the complete 46-character Marathi alphabet (Varnamala) and numerical sign set.
- Quantize the trained YOLOv8 model into TensorRT or ONNX formats to achieve 60+ FPS on embedded mobile edge processors.

- Implement a hybrid architecture combining YOLOv8 bounding box region extraction with a lightweight temporal network (e.g., Bi-LSTM) for continuous, moving gesture translation.

16. References

[1] C. Jochko et al., "YOLOv8: A Brief Review," arXiv preprint arXiv:2304.00501, 2023.

[2] C. Lugaresi et al., "MediaPipe: A Framework for Building Perception Pipelines," arXiv preprint arXiv:1906.08172, 2019.